Ujjwal Basumatary

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Education

Indian Institute of Science, Bangalore — **Nov 2020 – Apr 2025**
Bachelor and Master of Science, Major in Physics — *Physics: 9.6/10 (Overall: 9.3/10)*

Publications

- 2410.22702 — *Phys. Rev. D 111, L041306*
  Beyond Hawking evaporation of black holes formed by dark matter in compact stars, Ujjwal Basumatary, Nirmal Raj and Anupam Ray.

Preprints

- 2603.26090
  Cosmological Correlators Using Tensor Networks, Ujjwal Basumatary, Aninda Sinha and Xinan Zhou.
- 2512.02706
  Lectures on Quantum Field Theory on a Quantum Computer, Aninda Sinha and Ujjwal Basumatary.

Employment

Indian Institute of Science, Bangalore — **Jun 2025 – Jul 2025**
Project Associate - 1

Raman Research Institute, Bangalore — **Aug 2025 – Jun 2026**
Project intern — *National Quantum Mission*

Teaching assistantships

HE381: Quantum Field Theory and Quantum Computing — **IISc, Bangalore**
Instructor: Prof. Aninda Sinha — *CHEP*

- Conducted tutorials and designed/graded assignments, with demonstrations ranging from basic quantum algorithms to advanced Matrix Product States (MPS) algorithms.
- The implementation details of the quantum and MPS algorithms are in this GitHub repository.

Fellowships and Awards

S. N. Bhatt Memorial Fellowship — **May 2023 – Jul 2023**
International Center for Theoretical Sciences (ICTS) — *Bangalore*

Kishore Vaiyanik Protsahan Yojana (KVPY) Fellowship **Nov 2020 – Apr 2025**
Department of Science and Technology (DST) *India*

National Talent Search Exam (NTSE) Scholarship **Apr 2018 – Mar 2020**
National Council of Educational Research & Training (NCERT) *India*

Skills

- **Programming:** Julia, Python, Mathematica, C, C++ and Haskell.
- **Markup**: Proficient in LaTeX, working knowledge of HTML and CSS

Advanced Graduate Coursework

Non-eq. Many-Body Dynamics	10/10	AdS_3/CFT_2	9/10
Standard Model	10/10	Adv. Mathematical Physics	10/10
QFT - 2	9/10	Cosmology for Theorists	9/10
Statistical Field Theory	9/10	QFT - 1	10/10
General Relativity	10/10	Nuclear and Particle Physics	10/10